Sentiment analysis for stock market research: A bibliometric study[☆]Xieling Chen^a, Haoran Xie^b, Zongxi Li^b, Han Zhang^d, Xiaohui Tao^e, Fu Lee Wang^{c,*}^a School of Education, Guangzhou University, Guangzhou, China^b School of Data Science, Lingnan University, Hong Kong Special Administrative Region^c School of Science and Technology, Hong Kong Metropolitan University, Hong Kong Special Administrative Region^d Department of Finance, The Chinese University of Hong Kong, Hong Kong Special Administrative Region^e School of Mathematics, Physics, and Computing, University of Southern Queensland, Australia

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ABSTRACT

Sentiment analysis is widely utilized in stock market research. To comprehensively review the field, a bibliometric analysis was performed on 223 articles relating to sentiment analysis for stock markets from 2010 to 2022 collected from Web of Science database. Specifically, we recognized active affiliations, countries/regions, publication sources, and subject areas, identified top cited research articles, visualized scientific collaborations among authors, affiliations, and countries/regions, and revealed main research topics. Findings indicate that computer science journals are active in publishing works on sentiment analysis-facilitated stock market research. The research on sentiment analysis-facilitated stock market has attracted researchers from a wide geographic distribution, who have made significant contributions. The intensity of intra-regional collaborations is higher than that of inter-regional collaborations. Thematic topics regarding stock market research using sentiment analysis were detected using keyword mapping, with the following research topics being widely concerned by scholars: deep learning for stock market prediction, financial news sentiment empowered stock trend forecasting, effects of investor sentiment on financial market, and microblog sentiment classification for market prediction. Findings are helpful in depicting research status to researchers and practitioners, raising their awareness of research frontiers when planning research projects concerning sentiment analysis's application in stock markets.

1. Introduction

The close research on improving stock price prediction with the help of sentiment analysis technologies has increasingly received attention within academia. There are reviews of stock market prediction and its relevant topics (e.g., Thakkar and Chaudhari, 2021; Kumbure et al., 2022; Cheng et al., 2022; Das et al., 2024; Janková, 2023); however, none of them was conducted using quantitative methodologies. Bibliometrics involves the use of quantitative approaches for analyzing measurement of sizable textual data (Zupic and Čater, 2015). Based on bibliometrics, the present study aims to provide a comprehensive examination of the internationally academic literature regrading sentiment analysis for stock price prediction.

1.1. Stock price prediction and investor sentiments

According to the Efficient market hypothesis (EMH) (Fama, 1970), asset prices reflect all available information at any given time, thus it is

impossible to constantly attain higher returns than the market. While the EMH has been a cornerstone of financial theory, criticism about EMH and rational pricing of assets is increasing (Kumbure et al., 2022), arguing that markets are not always efficient according to behavioral finance (Shefrin, 2002) and that psychological factors and irrational behavior can cause mispricings. In line with this criticism, researchers have shown interest in applying sentiment analysis to capture market trends influenced by investor sentiments. In this study, sentiment refers to the attitudes, opinions, and emotions expressed by market participants and the general public toward financial markets, stocks, or economic conditions. These sentiments can be positive, negative, or neutral and are not merely derived from textual data like news articles and social media posts but encompass broader market psychology and behavioral factors.

Researchers have reported that anomalies such as momentum and mean reversion contradict the notion of market efficiency (Woo et al., 2020), indicating that historical price trends and patterns, as well as in-

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vestor sentiment, provide valuable predictive signals. As a result, stock investors and scholars have increasingly explored the possibility of predicting market movements based on stock historical data (Wu et al., 2022) using statistical time-series analysis approaches (Rahimi and Khashei, 2018) such as auto-regressive conditional heteroscedasticity models (Yilmaz, 2020) that focus on structured data analysis.

As social networks become increasingly popular, investors are inclined to show their viewpoints on stock investment, which has the potential to influence other investors' moods and thus their investment decision-making. For example, comments posted on social media platforms can generate sentiments or reactions in investors that can further impact their stock investment intentions (e.g., Li et al., 2014a; Chen et al., 2016; Alostad and Davulcu, 2017; Jiang et al., 2020), and financial news can also cause changes in stock market indices (Allcott and Gentzkow, 2017). Researchers have reported correlations between stock prices and investor sentiments expressed on social networks (Gan et al., 2020; Valle-Cruz et al., 2022) based on sentiment analysis. As a result, sentiment analysis has been valuable in stock prediction, especially with the advances in big data and artificial intelligence (AI), which contributes to sophisticated and effective models or systems for predicting stock market movements accurately (Kumbure et al., 2022).

1.2. Sentiment analysis for stock price prediction

According to Nabipour et al. (2020), public sentiments on social media contribute to accurately predicting stock market movements. Financial sentiment analysis has thus increasingly been considered useful for helping investors understand relationships among sentiments hidden within social media data or financial news with stock market prices (Mehta et al., 2021). Researchers have leveraged various sentiment analysis methodologies, including sentiment dictionaries, lexicon-based methods, machine learning, deep learning, and hybrid methodologies to interpret market sentiment from textual data.

First, sentiment dictionaries like Harvard Psychological and Loughran-McDonald, by providing pre-defined lists of words associated with positive or negative sentiments, are straightforward to use and can quickly identify sentiment in text based on word presence. However, they are often limited by their incapability to cover all relevant terms and understand context or detect sentiment in more complex language structures, thus reducing their accuracy in nuanced scenarios (Li et al., 2021).

Second, lexicon-based methods expand on sentiment dictionaries by incorporating more sophisticated rules and heuristics to consider the intensity of sentiment and handle some contextual nuances to determine sentiment (Shang et al., 2023). However, these methods share the same fundamental limitations as sentiment dictionaries, such as the reliance on pre-defined word lists and potential struggles with context-specific meanings.

Third, machine learning methodologies like support vector machines (SVMs) and decision trees, capable of detecting more complicated patterns and relationships within the data compared to lexicon-based approaches, are popularly utilized for sentiment prediction. Wu et al. (2021) adopted dimensional value-arousal for defining sentiments and trading intensity of stock market tendencies to promote predictions of stock market sentiments. Parray et al. (2020) utilized SVMs and logistic regression for predicting market indices. However, they are limited to the high requirements for substantial amounts of labeled data for training alongside a risk of overfitting.

Furthermore, deep learning approaches like convolutional neural networks (CNNs) and long short-term memory networks (LSTMs), particularly effective at capturing long-term dependencies and understanding complex language structures, represent the cutting edge of sentiment analysis (e.g., Zhao and Yang, 2023; Albahli and Nazir, 2024). For example, Zolfaghari and Gholami (2021) combined adaptive wavelet transformers, LSTMs, and ARIMAX-GARCH approaches for stock index predictions. Singh (2023) performed time series prediction using LSTMs

to increase stock price prediction accuracy. A post-training model on bidirectional encoder representations from transformers (BERTs) is reported to effectively extract aspects and classify sentiments (Xu et al., 2019). However, similar to machine learning, deep learning models also require significant resources for training with a need for large datasets to avoid overfitting and generalize to new data.

In recent years, hybrid approaches that combine elements from multiple methodologies to leverage their respective strengths while mitigating their weaknesses are increasingly adopted to improve stock market prediction accuracy. For instance, Jing et al. (2021) proposed a hybrid method where a CNN is first utilized to classify investors' concealed sentiments derived from a prominent stock forum, followed by the application of LSTMs to analyze technical indicators from the stock market alongside the sentiment analysis outcomes. Sharaf et al. (2023) introduced a hybrid system that integrates machine learning and deep learning models for mining social media data in conjunction with historical stock data to enhance overall stock trend prediction accuracy. These hybrid approaches, although can generally enhance accuracy and robustness, often come with increased complexity in implementation and tuning and require careful integration and balancing of different methods to ensure optimal model performance.

1.3. Reviews on stock market prediction

Several reviews regarding stock market prediction are available. For example, by systematically analyzing 106 studies related to stock market response to Internet information diffusion, Agarwal et al. (2019) revealed a lack of research regarding the influence of Internet information on stock markets and investor behaviors. Bustos and Pomares-Quimbaya (2020) systematically analyzed 52 stock market prediction articles published between 2014 and 2018 to understand prediction technologies for stock markets, finding (1) technical indicators' prevalence for forecasting stock markets, (2) sentiments' prevalence in social networking data analysis, and (3) increased interest in ensemble learning and deep learning for promoting prediction accuracy. Thakkar and Chaudhari (2021) first introduced various deep neural networks (DNNs) for prediction tasks and then exploited DNNs' effectiveness in forecasting stock prices/trends. By analyzing 138 studies from 2000 to 2019, Kumbure et al. (2022) examined markets and stock indices and 2173 unique variables (e.g., technical and fundamental indicators) utilized for predicting stock markets and introduced various machine learning empowered predictive approaches, finding an increased interest in DNNs especially LSTM networks for robust stock market predictions.

In terms of reviews on sentiment analysis for stock market prediction, Cheng et al. (2020) analyzed 9 articles about the sentiment-based stock forecasting approach to summarize sentiment analysis methods used in this field, as well as evaluating their advantages and disadvantages. Das et al. (2024) examined 76 articles on the use of graph neural networks (GNNs) combined with sentiment analysis for stock prediction, outlining the basic principles of GNNs and sentiment analysis, highlighting their contributions to the field of stock prediction, and pointing out the shortcomings of traditional methods. Based on 49 articles on text mining and sentiment analysis for stock market prediction, Janková (2023) evaluated the techniques employed for extracting sentiment from textual data, with particular attention to the generalizability and transferability of the research findings.

Based on the literature reviews, it is crucial to recognize that there are still key problems regarding sentiment analysis for stock market prediction that have not been adequately addressed in the literature. For instance, what are the primary areas of focus and research interests? Who are the productive institutions in the field? How do institutions and countries/regions collaborate? To answer these questions, researchers must be open to adopting novel methodologies, particularly from a quantitative perspective, to tackle these challenges.

1.4. Research aims and questions

The present study aims to broaden knowledge in the field of sentiment analysis for stock market predictions based on a bibliometric analysis and knowledge mapping of relevant academic literature from 2010 to 2022. This study differs from previous reviews in three ways. First, unlike earlier reviews that relied on qualitative and systematic analysis methods with a smaller set of papers being analyzed, this study adopted a quantitative analysis approach, thus findings and insights obtained are not limited to pre-established codes or categories. Second, identifying main sources and contributors helps researchers pinpoint international sources focused on sentiment analysis for stock market prediction, find suitable channels for contributing, and recognize leading contributors from whom they can gain knowledge. Additionally, this study offers a clear picture of the social structure of scientific collaborations among contributors, facilitating the identification of potential academic collaborators. Our analytical framework, which combines bibliometrics, social network visualization, and keyword mapping, aims at mining large-scale data instead of focusing on manual analysis of pre-defined dimensions of a small set of studies based on systematic analysis (e.g., a systematic review on theories and models underpinning sentiment analysis in stock markets). The bibliometric analysis has been popularly employed in diverse fields and has shown effectiveness in “understanding the historical and extant research progress, the development of technologies applied, and the drivers of fresh ideas” (Chen et al., 2022, p. 29). Specifically, the study addresses four research questions (RQs).

RQ1: What were the active sources, subject areas, countries/regions, and affiliations?

RQ2: What were the top-cited research articles?

RQ3: What do the scientific collaborations look like?

RQ4: What are the predominant research topics?

These RQs are determined by referencing previous bibliometric studies that, like the current study, aim to comprehend the research status of a specific field. Examples include automatic summarization (Chen et al., 2024a,b,c,d), sentiment analysis (Chen and Xie, 2020), sensitivity analysis in multi-criteria decision-making (Demir et al., 2024), AI in fake review detection (Jabeur et al., 2023), and digital innovation (Cheng et al., 2023). Answering these questions provides a state-of-the-art understanding of research on sentiment analysis for stock market prediction and offers valuable insights to researchers and project initiators for its future development (Donthu et al., 2020).

The contributions of this study include four aspects. First, answering RQ1 enables researchers to identify appropriate platforms for sharing and publishing studies on sentiment analysis for stock market prediction and to recognize leading contributors from whom they can learn (Song et al., 2019). Second, tackling RQ2 helps in understanding how sentiment analysis can be effectively integrated to enhance stock market prediction and applies findings from influential research (Chen and Xie, 2020). Third, addressing RQ3 aids in understanding collaboration connections and finding potential academic collaborators (Matveeva et al., 2022). Moreover, answering RQ4 keeps researchers and practitioners aware of crucial issues that require attention when undertaking scientific or technological endeavors related to sentiment analysis for stock market prediction (Maryanti et al., 2023). Additionally, the analytic framework with the combination of bibliometrics, social network visualization, and keyword mapping, is domain-independent and can be implemented to various areas beyond stock market analysis to process and analyze large-scale literature textual data to uncover patterns and relationships that enlighten studies across disciplines.

2. Methods

This study followed Preferred Reporting Items for Systematic Reviews and Meta-Analyses Flow Diagram (Moher et al., 2009) for the identification, selection, and critical analysis of relevant papers using bibliometrics, social network visualization, and keyword mapping, as described in the following sections.

2.1. Data retrieval and selection

We referred to previous review studies on sentiment analysis and stock market to identify relevant search terms (e.g., Agarwal et al., 2019; Bustos and Pomares-Quimbaya, 2020; Thakkar and Chaudhari, 2021; Kumbure et al., 2022; Chen and Xie, 2020), which were used to search in the Web of Science (WoS) database to match titles, abstracts, and keywords. The WoS, as the world's most widely used database that contains publications from impactful venues, has been popularly adopted in review studies (e.g., Chen et al., 2024a,b). An article was collected if it was: (1) a journal article or conference paper, (2) written in English, and (3) published from 2010 to 2022. Accordingly, 296 articles were identified. Fig. 1 presents the data search and selection procedure.

In this study, the data screening process was performed by two domain experts, specifically the second and third authors, who have extensive experience in researching sentiment analysis for stock market prediction. Both experts hold advanced degrees in fields related to data science, with a focus on sentiment analysis, and each expert has published multiple peer-reviewed articles on sentiment analysis and its application in stock market prediction.

During the screening, the two experts independently assessed a total of 296 articles according to the exclusion and inclusion criteria integrated in Fig. 1 to identify those relevant to sentiment analysis for stock market prediction. The inter-rater reliability, measured using Cohen's kappa, was 94%, indicating a high level of agreement between the experts, with any inconsistencies being discussed and resolved through detailed discussions to ensure that the final decisions were well-founded and mutually agreed upon. In addition, in cases where consensus could not be reached, the corresponding author, a third expert with substantial experience in the field, was involved to review and re-evaluate the disputed articles and contributed to the final decision-making process. To evaluate the performance of the domain experts, prior to the screening process, the experts participated in training sessions to calibrate their understanding and application of the exclusion and inclusion criteria to minimize discrepancies and align their assessments. Furthermore, throughout the screening process, regular cross-checks were conducted to monitor consistency and identify any emerging patterns of disagreement to allow for timely adjustments and refinements in the screening. In addition, the experts provided feedback on the screening criteria during the screening, which was used to continuously improve the rigor and clarity of the selection criteria.

Accordingly, 73 articles were excluded, including (1) 18 studies focusing only on stock market, (2) 35 studies focusing only on sentiment analysis, (3) 16 studies not focusing on stock market or sentiment analysis, (4) 2 studies not reporting results, (5) 2 editorials or opinion papers, and (6) 2 studies without full-texts. Finally, 223 articles were utilized for data analysis, including (1) 136 articles about stock market analysis with sentiment as a main focus, (2) 47 articles about stock market research using sentiment analysis combined with methods such as technical analysis, (3) 33 articles about sentiment analysis in stock-related contexts, and (4) 7 survey studies on sentiment analysis's usage in the stock market. Fig. 2 depicts the number of the included articles by year, which reflects the growing interest in the application of sentiment analysis in stock market research.

2.2. Data analysis

We analyzed the 223 articles from the perspectives of contributors, collaborations, and thematic structures using bibliometric indices, social network visualization, and keyword mapping. Regarding RQ1, this study utilized the number of articles/citations and the Hirsch index (H-index) to identify active sources, subject areas, countries/regions, and affiliations. For RQ2, we followed Chen et al. (2023) to consider the citation count of each article (Chen et al., 2022) by calculating annual citations (C/Y) using $C/Y = \text{total citations} / (2022 + 1 - \text{publishing year})$.

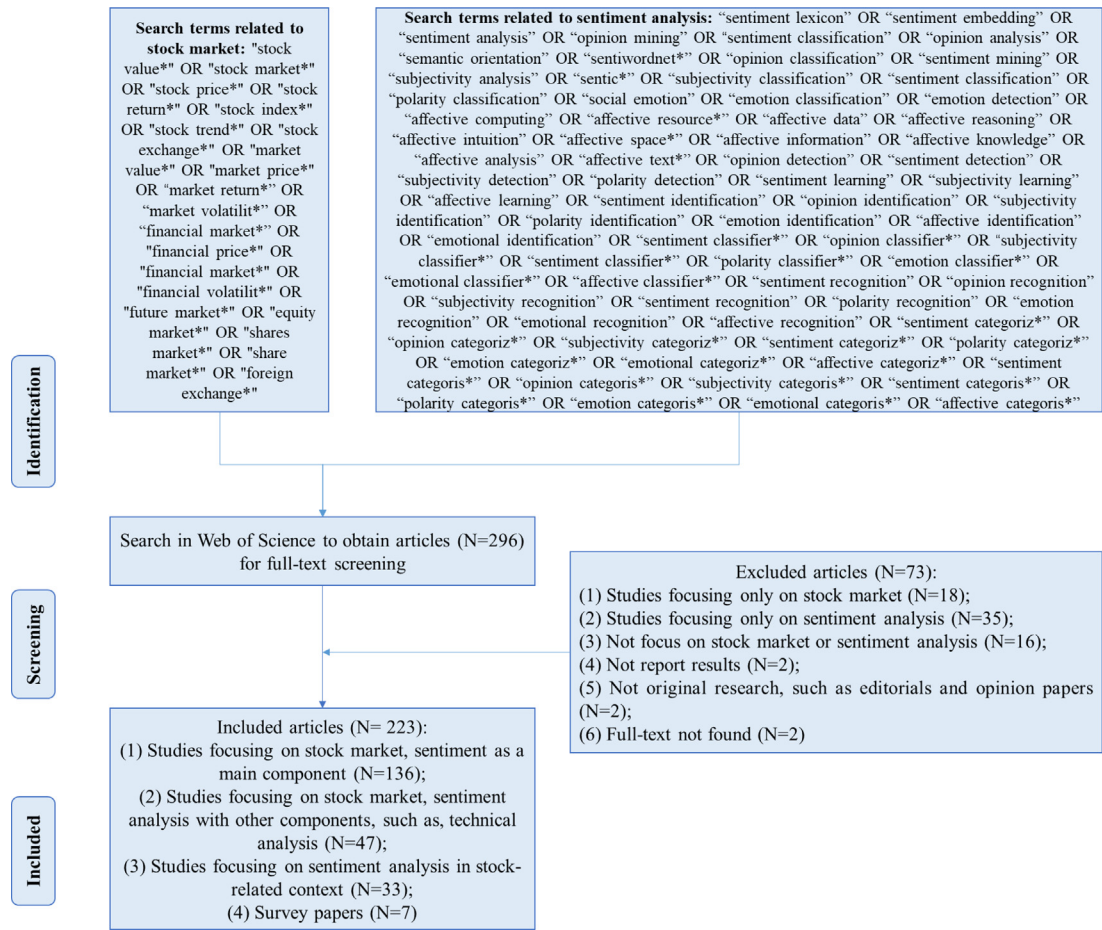


Fig. 1. Data retrieval and selection.

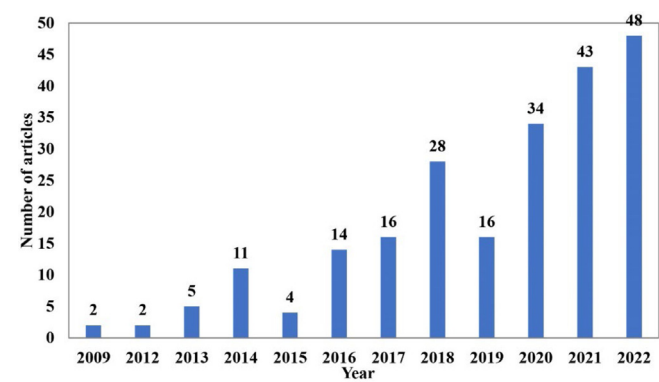


Fig. 2. Annual distribution of the 223 articles.

to identify top 10 research articles.

Regarding RQ3, social network visualization using Gephi¹ was adopted to depict collaborations. Specifically, we selected an undirected graph type, meaning the links between nodes do not have arrows indicating direction. Nodes were color-coded based on the region of the corresponding institution or country. We used the Fruchterman Reingold layout in Gephi 0.1.0 to analyze the collaboration networks of authors, institutions, and countries by measuring the distance between nodes using a combination of attractive and repulsive forces based

on a spring-electrical model (Zhong et al., 2023). Node sizes were determined by the publication volume of the corresponding institution or country, with sizes constrained within a range of 20 to 80 units. The thickness of the links was determined by the number of collaborations between the two nodes.

For RQ4, we extracted from author-defined keywords, keywords provided by WoS, abstracts, and titles and computed and ranked their frequencies to identify reoccurring topics (Chen et al., 2021), which were visualized as a keyword co-occurrence map using VOSviewer.² Specifically, we set a minimum occurrence threshold by including only keywords with a minimal frequency of 7 in the final map, with the strength of the co-occurrence links between keywords being normalized using the association strength method. We set the attraction and repulsion parameters as 2 and 1, respectively to adjust the layout of the network. In terms of network visualization, VOSviewer employs a VOS clustering algorithm to identify groups of closely related keywords. The clustering resolution parameter was set to 1.0 to balance the size and number of clusters, with a minimum cluster size of 1 to indicate at least one cluster should be included. The resulting bibliometric map was visualized with nodes representing keywords and edges representing co-occurrence links. The size of the nodes indicates the frequency of the keywords, while the thickness of the edges reflects the strength of the co-occurrence links, with different colors being used to distinguish between clusters.

¹ <https://gephi.org/>

² <https://www.vosviewer.com/>

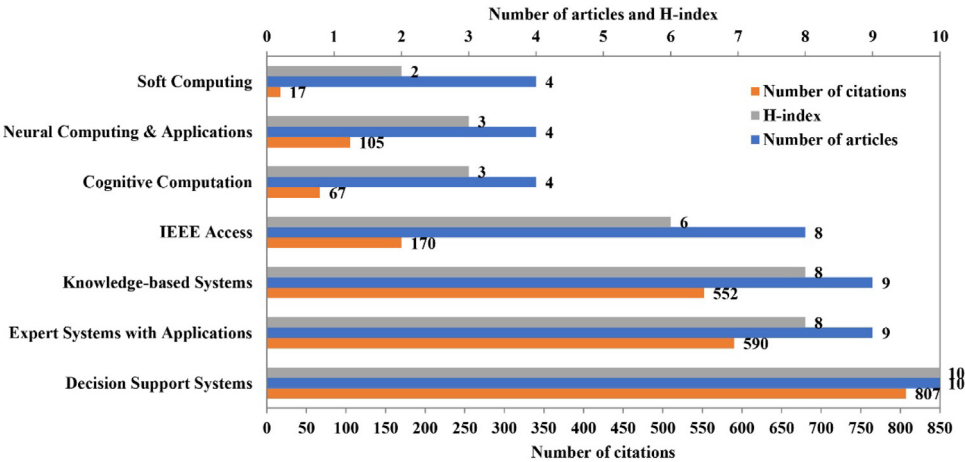


Fig. 3. Top journals.

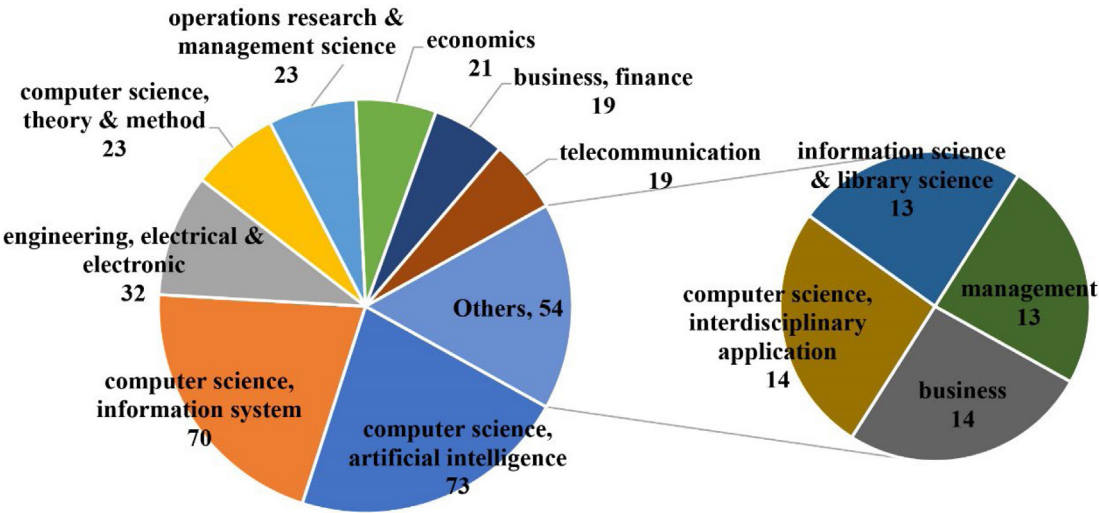


Fig. 4. Top subject areas.

3. Results

The bibliometric analysis results of the 223 articles on sentiment analysis for stock market prediction are presented, including the analysis of publication sources, subject areas, countries/regions, and affiliations, top cited research articles, scientific collaboration analysis, as well as the analysis of predominant research topics.

3.1. Publication sources, subject areas, countries/regions, and affiliations (RQ1)

There are 144 sources contributing to the publication of 223 articles on sentiment analysis for stock market prediction from 2010 to 2022; the top 7 productive journals listed in Fig. 3 together accounted for 21.52% of the analyzed articles. The top 3 sources from the perspectives of article count, the H-index, and citation count are Decision Support Systems, Expert Systems with Applications, and Knowledge-based Systems.

A total of 43 WoS categories are found to be related to research on sentiment analysis in the stock market, with the top 12 categories ranked by article count being shown in Fig. 4. Among them, the top 2 are computer science, artificial intelligence, and information system, followed by engineering, electrical & electronic, computer science theory & method, operations research & management science, economics, business, and finance.

There are 55 countries/regions contributing to the publication of 223 articles; the top 8 productive countries/regions are listed in Fig. 5. The top 3 are China, the USA, and India, with China publishing 52 articles, taking up a proportion of 23.32%; ranked by H-index, the USA (18), China (17), and Taiwan (10) are the top 3 countries/regions.

There are 346 affiliations contributing to the publication of 223 articles from 2010 to 2022; the top 6 productive affiliations are listed in Fig. 6. The top 5 are Chinese Academy of Sciences, City University of Hong Kong, Nanyang Technological University, Renmin University of China, and University of Chinese Academy of Sciences. Ranked by citation count, Nanyang Technological University (305), City University of Hong Kong (255), and Texas Tech University (292) are the top 3 affiliations in this research field.

3.2. Top cited research articles (RQ2)

The top articles ranked by annual citations are presented in Table 1. The 10 articles can be grouped into three types from the perspective of data used for sentiment analysis, including social media data (Nguyen et al., 2015; Jin et al., 2020; Valle-Cruz et al., 2022; Sohangir et al., 2018), financial news articles (Li et al., 2014, 2020; Picasso et al., 2019; Weng et al., 2018), and combined use of social media and financial news data (Yu et al., 2013; Audrino et al., 2020).

Studies using social media data for sentiment analysis included Nguyen et al. (2015), Jin et al. (2020), Valle-Cruz et al. (2022), and

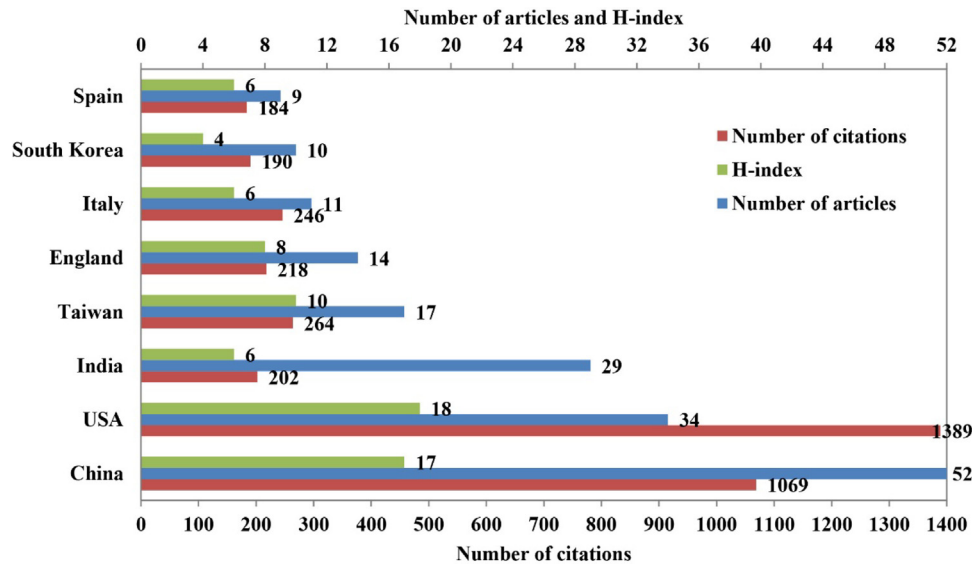


Fig. 5. Top countries/regions.

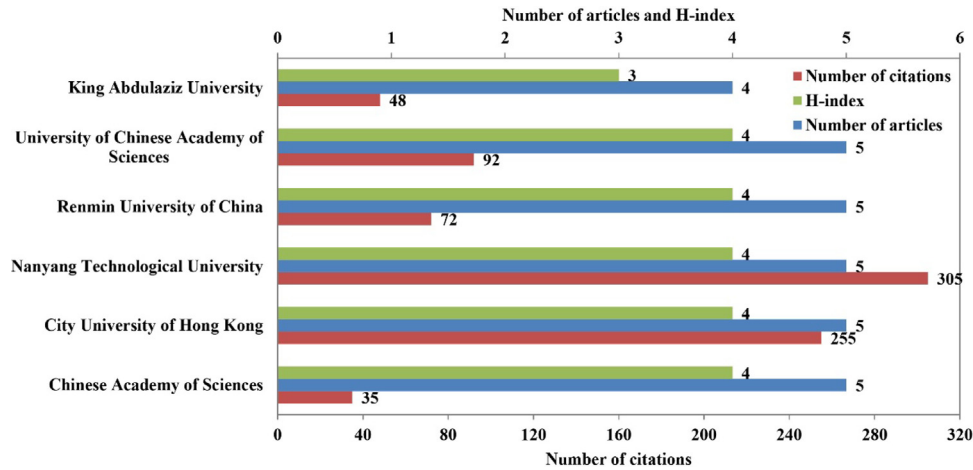


Fig. 6. Top affiliations.

Sohangir et al. (2018). Nguyen et al. (2015) used joint sentiment-topic methods and SentiWordNet to extract topics and their associated sentiments from social media texts and then employed them for predicting stock movements, achieving 2.07% better performance than models using only historical prices. Nguyen et al. introduced a novel “topic-sentiment” feature to enhance stock market prediction performance and conducted a large-scale experiment to investigate the integration of sentiments; however, they merely considered historical prices and sentiments expressed on social media. Jin et al. (2020) combined CNNs with word2vec to categorize stock market comments into bearish and bullish groups and calculated a daily sentiment index to forecast stock closing prices using LSTM, with experiments on the AAPL dataset showing that their approach increased prediction accuracy and decreased time delay. Jin et al.’s contribution lies in leveraging a large volume of stock market comments that reflect overall investor sentiment and integrating investor sentiments into stock prediction to increase accuracy; however, because they used CNN as the base model for the sentiment index, high-quality stock review datasets are required in their study.

Valle-Cruz et al. (2022) explored Twitter post polarity’s effects on financial indices during pandemics by analyzing the sentiment of tweets from influential Twitter accounts, using a lexicon-based approach to correlate tweet polarities with financial market indicators.

The results indicated that their method was more effective in detecting highly shifted correlations using SenticNet (Cambria et al., 2020) compared to other lexicons like Bing Liu (Liu, 2012), Sentiment 140 (Kiritchenko et al., 2014), NRC (Mohammad and Turney, 2013), and Affin (Keshavarz and Abadeh, 2017). Valle-Cruz et al. contributed by improving sentiment analysis through the combination of a lexicon-based approach with shifted correlation analysis to reveal hidden correlations between Twitter post polarity and financial market behavior over an 11-year period. However, since their study focused on the influence of Twitter post polarity on financial indices during pandemics, the results may not be applicable to non-pandemic situations or other types of market events. Sohangir et al. (2018) adopted LSTMs, doc2vec, and CNNs to determine the sentiments of posts on StockTwits, demonstrating the effectiveness of deep learning for financial sentiment analysis, with CNNs showing the best prediction performance. Sohangir et al. contributed by showcasing the superiority of CNNs over traditional data mining methods in stock sentiment analysis; however, they focused solely on CNNs and may have overlooked the potential performance and applicability of other advanced deep learning approaches for sentiment analysis in stock market predictions.

Studies that have utilized financial news for sentiment analysis included Li et al. (2014), Picasso et al. (2019), Li et al. (2020), and Weng et al. (2018). Li et al. (2014) examined how different dimensions

Table 1
Top cited research articles.

Studies	Research purpose	Data	Sentiment analysis	Contributions	Main findings	Limitations	Year	C/Y
Nguyen et al. (2015)	stock movement prediction	social media data	joint sentiment/topic model, SentiWordNet	proposed a feature “topic-sentiment” to improve stock market prediction	the proposed method achieved 2.07% better performance than the model using historical prices only	only used historical prices and sentiments derived from social media	2015	27.38
Yu et al. (2013)	media’s effects on company equity values	social and conventional media data	Naïve Bayes	explored the combined impact of multiple social media sources and conventional media on firm performance	social media had a stronger influence on firm stock performance compared to conventional media	did not make specific adaptations to domain-specific contexts	2013	25.40
Li et al. (2014)	news’ effects on stock returns	financial news articles	Harvard Psychological and Loughran–McDonald dictionaries, SenticNet, and bag-of-words	demonstrate the effectiveness of integrating sentiment analysis into stock price prediction	the integration of sentiment analysis improved prediction accuracy across individual stocks, sectors, and indices compared to the bag-of-words model	did not address the challenge of analyzing intra-day stock price returns that could be complicated by higher noise levels	2014	24.78
Li et al. (2020)	stock market prediction	financial news articles	SenticNet 5, SentiWordNet 3.0, Vader and Loughran–McDonald financial dictionary 2018	developed a stock prediction system that combined technical indicators from stock prices with sentiment analysis of news articles based on LSTM and sentiment dictionaries	the proposed system consistently outperformed baselines in terms of prediction accuracy and F1 score across individual stocks and sectors	relied on static sentiment dictionaries for news sentiment analysis, which might not capture evolving sentiment dynamics	2020	24.00
Jin et al. (2020)	stock closing price prediction	stock market comments	CNNs and word2vec	exploited stock market comments that reflected the overall investors’ sentiment orientation for stock prediction	the proposed method not only improved prediction accuracy, but also reduced time delay.	high requirements for stock review datasets since CNNs were used as the basis model of the sentiment index	2020	23.33
Valle-Cruz et al. (2022)	Twitter’s effects on stock market decisions	influential Twitter account posts	lexicon-based approaches such as NRC, Affin, and SenticNet	combined a lexicon-based approach with shifted correlation analysis to uncover latent correlations between Twitter post polarity and financial market behaviors to enhance sentiment analysis	the proposed method was effective in detecting highly shifted correlations using SenticNet	results might not generalize to non-pandemic conditions or other types of market events	2022	23.00
Audrino et al. (2020)	effects of sentiment and attention variables on stock market volatility	social media data and financial news articles	multi-layer CNNs and Deep-MLSA	integrated sentiment and attention variables from diverse sources, alongside a high-dimensional predictive regression model for modeling volatility	sentiment and attention variables significantly enhanced predictive accuracy for future volatility	did not explore potential non-linear effects, heterogeneities across different companies and time periods	2020	22.33
Picasso et al. (2019)	market trend prediction	financial news articles	Loughran and McDonald dictionary	highlighted the potential of integrating sentiment with technical indicators to enhance stock market predictions	sentiment embeddings from AffectiveSpace 2 and McDonald achieved high accuracy and annualized returns, respectively	could not capture evolving sentiment contexts in news articles	2019	22.25
Sohangir et al. (2018)	financial sentiment analysis	Stock Twits posts	LSTMs, doc2vec, and CNNs	demonstrated the potential of deep learning for financial sentiment analysis	CNNs improved sentiment prediction accuracy for StockTwits posts	overlook the applicability of other deep learning models for sentiment analysis in stock market predictions	2018	19.40
Weng et al. (2018)	short-term stock price prediction	financial news articles	FinSentS web news sentiment	developed a system that integrated diverse data sources and utilized ensemble methods for stock price prediction	the proposed system achieved a higher mean absolute percent error than the reported results in the literature	did not fully capture variations in sentiment dynamics	2018	16.00

Note: C/Y: annual citations, calculated as total citations/(2022 + 1–publishing year).

of news sentiment affected stock price prediction by employing a framework that combined the Harvard Psychological and Loughran–McDonald financial sentiment dictionaries, SenticNet, and a bag-of-words approach for a quantitative assessment of sentiment in financial news articles. The experiments on five years of historical datasets from the Hong Kong stock exchange showed that sentiment analysis enhanced prediction accuracy for individual stocks, sectors, and indices compared to the bag-of-words model. Li et al. demonstrated the benefits of incorporating sentiment analysis into stock price prediction across different market levels; however, they did not address the challenge of analyzing intra-day stock price returns that may be affected by higher levels of noise. Picasso et al. (2019) integrated technical and fundamental analysis with machine learning to predict stock market trends, illustrating the effectiveness of sentiment analysis from news articles. They particularly emphasized the superior performance of sentiment embeddings from AffectiveSpace 2 in achieving high accuracy and the Loughran and McDonald dictionary in delivering higher annualized returns. Picasso et al. contributed by creating a robust model that successfully classified positive and negative trends within a stock portfolio and highlighted how combining sentiment with technical indicators improves stock market predictions. However, their reliance on pre-defined sentiment dictionaries such as Loughran and McDonald or AffectiveSpace 2 may limit the model's ability to capture nuanced or evolving sentiment contexts in news articles, potentially reducing its adaptability to changing market conditions.

Li et al. (2020) created a stock prediction system that combined technical indicators from historical price data with sentiment vectors from financial news articles and employed a two-layer LSTM neural network to account for sequential patterns in stock price trends. Experiments conducted on over five years of real data from the Hong Kong stock market demonstrated that their approach consistently outperforms baseline models in prediction accuracy and F1 score across different stocks and sectors, with the Loughran–McDonald Financial Dictionary showing the most significant improvement in modeling news sentiments. Li et al. contributed by developing a unified stock prediction system that merged technical indicators with sentiment analysis of news articles based on LSTMs and finance-specific sentiment dictionaries; however, their reliance on static sentiment dictionaries may not adequately address the nuanced or evolving sentiment dynamics in a rapidly changing market environment. Weng et al. (2018) designed a financial expert system using ensemble methods that integrated multiple data sources for extracting Google, Wikipedia, and financial news sentiment to forecast short-term stock prices. Experiments on Citi Group stock data revealed that their expert system achieves strong stock price predictions, with lower mean absolute percent error compared to results reported in previous literature. Weng et al. contributed by developing a unique financial expert system that combined various data sources and applied ensemble methods to improve stock price prediction accuracy; however, their focus on short-term predictions (1 to 10 days) may not fully account for variations in sentiment dynamics and other factors that influence longer-term stock movements.

Studies that utilized both social media and financial news data included Yu et al. (2013) and Audrino et al. (2020). Yu et al. (2013) employed an automated sentiment analysis technique to extract sentiment polarity at the sentence level from both social and conventional media and examined the comparative importance and interaction of these media types on the short-term stock market performance of firms. Experiments on 52,746 messages from 824 firms across different social and traditional media sources showed a stronger impact of social media on firm stock performance compared to conventional media. Yu et al. contributed by investigating the combined influence of multiple social media sources and traditional media on firm performance, underlining their relative importance and interactions; however, they did not make specific adaptations to domain-specific contexts, such as accounting or finance, which might lead to difficulties in capturing the nuanced sentiments relevant to financial markets accurately. Audrino et al.

(2020) extracted the sentiments from social media tweets and financial news articles using multi-layer CNNs and deep-MLSA and explored how sentiment and attention variables from these sources, along with search engine data, affect daily stock market volatility. Their experiments on New York stock data demonstrated that sentiment and attention variables significantly improved predictive accuracy for future volatility, especially for companies with large market capitalizations and lower shares held by institutional investors. Audrino et al. contributed by integrating a broad range of sentiment and attention variables from various sources using advanced sentiment analysis techniques, while also accounting for economic and financial variables in a high-dimensional predictive regression model suited for modeling volatility. However, their focus on linear relationships between sentiment variables and volatility does not address potential non-linear effects, variations across different companies and time periods, or differences in impact across various types of investors.

3.3. Scientific collaborations (RQ3)

The collaborations among 10 countries/regions, 4 institutions, and 4 authors are presented (Fig. 7), with the number of papers that two countries/regions, institutions, or authors collaborated not less than 3. Regarding countries/regions (Fig. 7(a)), there are two groups of collaborators, including (1) China, the USA, Australia, Canada, and Hong Kong, and (2) India, England, Saudi Arabia, Pakistan, and Egypt. The USA and China collaborated in publishing 9 articles, followed by Saudi Arabia and England, Canada and China, and Hong Kong and China, each of which collaborated on five articles. Regarding institutions (Fig. 7(b)), there are two groups of collaborative partners, including (1) Chinese Academy of Sciences and University of Chinese Academy of Sciences, and (2) University of Campinas and Instituto Federal de Sao Paulo. Regarding authors (Fig. 7(c)), there are two groups of collaborative partners, including (1) Erik Cambria and Frank Z. Xing, and (2) Arthur Emanuel de Oliveira Carosia and Guilherme Palermo Coelho.

3.4. Predominant research topics (RQ4)

The co-occurrence of 104 keywords used in the 223 articles are presented in Fig. 8, displaying 5 clusters. The keywords in the blue cluster mainly dealt with social media sentiment analysis to understand stock markets. The cluster in red more specifically addresses investor sentiment's effect on the financial market. The purple color-marked cluster has a focus on news text and opinion mining to understand the stock market movement. Keywords involved in the yellow cluster were more specifically related to news sentiment classification for market prediction. The last cluster in green color primarily involves keywords related to experimental studies on stock price predictions using DNNs.

4. Discussion

Based on 223 research articles retrieved from the WoS, this study focuses on the status and thematic structures of the field of sentiment analysis for stock market prediction.

4.1. In response to RQs

In response to RQ1, according to Figs. 3 and 4, at present, most articles on sentiment analysis for stock market prediction are published in Elsevier journals (e.g., Decision Support Systems, Expert Systems with Applications, and Knowledge-based Systems) in the field of computer science, artificial intelligence, and information system. Regarding country/region analysis (see Fig. 5), the growth of the number of articles is mainly contributed by researchers from countries/regions such as China and the USA. Results of the institution analysis (see Fig. 6) further demonstrate the active contributions from Asian affiliations

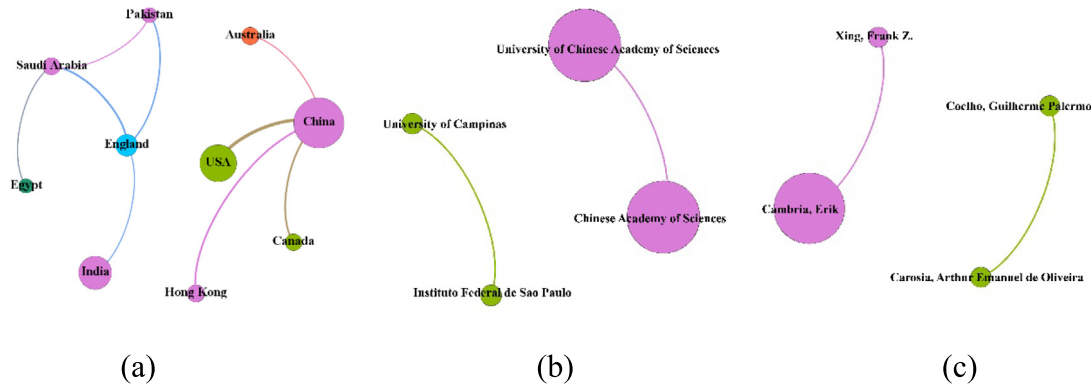


Fig. 7. Scientific collaborations among (a) countries/regions; (b) affiliations, and (c) authors (collaborative time ≥ 3).

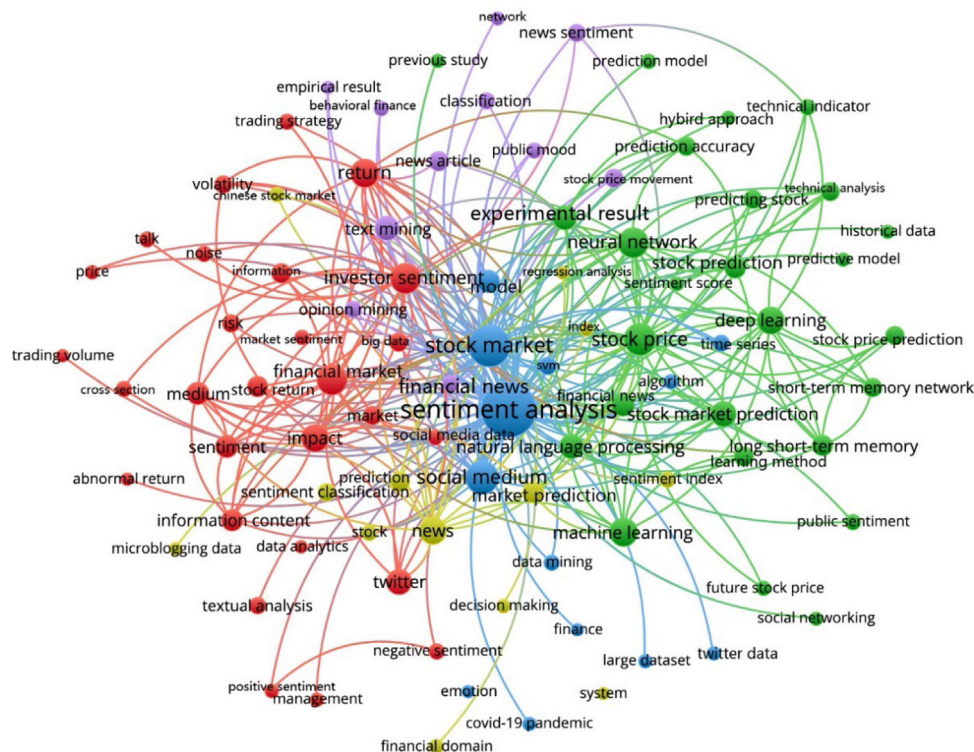


Fig. 8. Keyword co-occurrence.. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

(e.g., Chinese Academy of Sciences, City University of Hong Kong, Nanyang Technological University, Renmin University of China, and University of Chinese Academy of Sciences). These results corroborate with [Kumbure et al. \(2022\)](#) who have also reported the significant role of Asian affiliations in research on machine learning for stock market prediction.

In response to RQ2, according to [Table 1](#), the analysis of the top 10 research studies reveals several common themes and significant implications for the field, underscoring the potential of integrating different data sources, for example, social media, financial news articles, and conventional media, to capture investor sentiment and predict stock market movements. The use of advanced sentiment analysis tools, including sentiment dictionaries and deep learning models, also highlights the importance of leveraging sophisticated methodologies to enhance prediction accuracy.

In response to RQ3, according to Fig. 7 (a), countries and regions with higher levels of international collaboration commonly experience rapid development, as demonstrated by the USA and China, both of which are identified to be productive in the field (see Fig. 5). Figs.

7(b) and (c) suggest that institutions and authors within the same countries or regions often collaborate, aligning with previous studies (e.g., Chen et al. 2024; 2023) that have reported the prevalence of nearest-neighbor collaboration in scientific research, which may be a result of the ease of research exchange and resource sharing. However, Guerrero Bote et al. (2013) suggest mitigating negative local impacts by forming partnerships with distant countries or regions to enhance influence; thus, an increase in cross-regional collaborations within the sentiment analysis for the stock market prediction community is encouraged.

In response to RQ4, according to Fig. 8, thematic features relating to research on sentiment analysis for stock market prediction were revealed by keyword mapping represented by distinct clusters, highlighting four interconnected themes, including deep learning for stock price prediction, stock trend prediction based on news sentiments, investor sentiments' effects on financial market, and social media sentiment analysis for stock trend prediction. The four themes demonstrate the importance of integrating diverse data sources, for example, social media, news articles, and traditional sentiment indicators, and

the interplay between different types of sentiment data, for example, social media sentiment, investor sentiment, and news sentiment for developing robust models that reflect the multifaceted nature of market sentiment.

4.2. Current status and future directions

The landscape of sentiment analysis for stock market prediction is increasingly rich and multifaceted, encompassing a range of data sources, analytical techniques, and thematic focuses identified by a joint analysis of top 10 research studies and keyword mapping results.

The first theme focuses on the increasingly prevalent use of deep learning models such as CNNs and LSTMs for stock price prediction, driven mainly by their capabilities in representation learning to transform raw data for the purpose of learning complicated, multiple-level patterns (Latif et al., 2020). In Chandar (2022), ten technical indicators extracted from historical data were transformed into images using Gramian Angular Field and integrated into CNNs as feature vectors to improve prediction accuracy. Daradkeh (2022) developed a hybrid CNN-BiLSTM model for stock trend prediction, with experiments on Dubai Financial Market data demonstrating improved stock market prediction accuracy by jointly using news events/sentiments and quantitative financial data.

The second theme reveals the prevalence of using news sentiments extracted from financial news articles that reflect the information flow influencing investor behavior to improve stock trend prediction. In the big data era, with a tremendous public data being increasingly accessible (Rumsfeld et al., 2016), the need for automatic text classification has grown (Binkhounain and Zhao, 2019), driving the application of natural language processing (NLP), classification, and deep learning methodologies for news sentiment analysis to predict stock trends. For example, Sinha et al. (2022) extracted fine-grained financial sentiments for entities identified from news articles based on the SEntFiN 1.0 dataset, and Anbaee Farimani et al. (2022) developed a FinBERT-empowered predictive model for computing financial news headlines' sentiments.

The third theme focuses on understanding how investor sentiment influences financial markets. As social media becomes prevalent, interactions among stock investors are increasingly frequent (Jin et al., 2020), with sentiments expressed on social media gradually becoming crucial for influencing stock movements (Li et al., 2017). Scholars have used dictionary-driven approaches and machine learning to capture and examine online investor sentiments' associations with stock market movements. For example, Nguyen and Huynh (2022) mined textual reports based on sentiment dictionaries and rule-driven sentiment analysis to decide financial managers' attitudes (positive, negative, or neutral). Similarly, Chun et al. (2021) utilized semantic orientation from Laplace smoothed normalized pointwise mutual information to examine individual investors' sentiments and their relationship with stock returns/volatility, showing that negative sentiment was negatively related to market returns. Loginova et al. (2024) leveraged a joint/sentiment topic model and a topic sentiment latent Dirichlet allocation model to incorporate joint topical-sentiment features and text subjectivity levels for improving the prediction of directional bitcoin price returns.

The fourth theme reflects an increasing interest in exploiting sentiments expressed on social media platforms, particularly microblogs which have developed into a useful tool for people to post their feelings and opinions with regard to topics like politics, finance, and the economy in real time (Sun et al., 2020), for stock trend prediction. Given the increasing popularity of online investment communities, including stock microblogs among individual investors and financial professionals (Li et al., 2018), scholars have used sentiment analysis for the automatic mining of opinionated content in volume to recognize microbloggers' opinions (Oliveira et al., 2016). For example, Ye et al. (2021) focused on fusing information with varied features for

microblog sentiment classification using multi-view ensemble learning, and Sun et al. (2020) classified microblogging users into different clusters based on their identities and examined different clusters' effects on the stock market.

Findings of the top studies and keyword mapping reveal the increasing reliance on advanced computational techniques such as DNNs to process and analyze sentiment data, given their great potential in handling the complexity and volume of data generated by social media platforms and news outlets to enable researchers to extract and classify sentiment from large datasets with greater accuracy to making timely and informed market predictions. However, most studies such as Yu et al. (2013) have not tailored their models to domain-specific contexts, thus limiting the accuracy of sentiment analysis. Second, the reliance on static sentiment dictionaries (e.g., Li et al., 2020) may not accurately reflect the evolving nature of market sentiment. Furthermore, studies like Valle-Cruz et al. (2022) noted that their findings might not generalize to non-pandemic conditions or other types of market events. In addition, studies like Audrino et al. (2020) did not explore potential non-linear effects and heterogeneities across different companies and time periods.

To address these limitations and advance the field of sentiment analysis for stock market prediction, several future directions are suggested. The first is to develop models that adapt to real-time data and evolving market conditions, for example, integrating real-time data feeds from various sources and using machine learning techniques to update sentiment dictionaries dynamically. The second suggestion focuses on combining multiple data sources and hybrid models to provide a holistic view of market sentiment (e.g., Ma et al., 2023; Chen et al., 2024c). Another suggestion is to investigate the non-linear effects of sentiment on stock prices and examine heterogeneities across different market segments to provide deeper insights into market dynamics and capture the complex relationships between sentiment and market behavior. Finally, we suggest expanding the scope of research to include a wider range of market events and conditions to ensure the robustness and generalizability of sentiment analysis models.

5. Conclusion and implications

This study employed bibliometrics and knowledge mapping techniques to examine scientific literature regarding sentiment analysis for stock market prediction by pinpointing the primary contributors and illustrating their scientific partnerships, as well as depicting the research status and progress from 2010 to 2022 to increase researchers' awareness of the research frontiers for their research and project planning. Our analysis was based on records retrieved from the WOS. Although the WOS is a multidisciplinary database of academic output and are widely used for literature reviews, there might still be records regarding sentiment analysis for stock market prediction that were ignored. Future studies could incorporate additional data sources like Scopus. Moreover, future research could adopt more sophisticated analysis methods, such as topic modeling, to gain deeper insights into how sentiment analysis aids stock prediction. Nevertheless, as the first quantitative review of this field, the implications of this study included five aspects.

First, the rapid development and potential of sentiment analysis for stock market prediction provide numerous research opportunities for new researchers and Ph.D. students. Exploring various aspects of sentiment analysis for stock market prediction, such as innovative algorithms and applications in specific domains, particularly through interdisciplinary approaches, collaboration with experts from different fields, and integrating insights from multiple areas like NLP, machine learning, finance, and behavioral science, contribute to developing effective real-world financial sentiment analysis applications.

Second, international journals specializing in related fields (e.g., NLP, machine learning, and information retrieval) play a crucial role in promoting sentiment analysis for stock market prediction research.

It is also suggested that special issues can be launched to encourage researchers from diverse backgrounds to contribute their expertise and foster cross-disciplinary collaboration.

Third, scholars, especially newcomers, can consider the main contributors identified in this study as potential role models from whom they can learn and establish collaborations since regional and cross-regional collaborations are helpful for researchers to effectively explore opportunities and address the challenges in sentiment analysis for stock market prediction.

Furthermore, researchers are encouraged to continue focusing on topics that attract wide and growing interest, such as deep learning for stock price prediction, stock trend prediction based on news sentiments, investor sentiments' effects on financial markets, and social media sentiment analysis for stock trend prediction.

Additionally, as sentiment analysis for stock market prediction becomes increasingly integrated into practical applications, collaborations between academics and professionals from computer science, finance, and other relevant fields should be strengthened to facilitate the development of effective real-world financial sentiment analysis systems tailored to specific domains or use cases.

CRediT authorship contribution statement

Xieling Chen: Writing – original draft, Visualization, Methodology, Formal analysis, Conceptualization. **Haoran Xie:** Writing – review & editing, Investigation, Conceptualization. **Zongxi Li:** Methodology, Data curation. **Han Zhang:** Validation, Software. **Xiaohui Tao:** Visualization, Resources. **Fu Lee Wang:** Writing – review & editing, Supervision, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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